where $\mathscr{I}$ is a purely numerical constant. (Detailed calculations show that $\mathscr{I} = 2$ for $N_f = 2$ and $N_c = 3$.) Adding the terms (29.3.23) and (29.3.24), the complete effective superpotential is now

$$f_{\text{total}}(q, \overline{q}) = \mathscr{K} \left[\prod_i u_i\right]^{-2/(N_c - N_f)} + \sum_i m_i u_i^2 \,, \tag{29.3.29}$$

where

$$\mathscr{K} \equiv \mathscr{I} \exp\left(\frac{i\theta}{N_c - N_f}\right) \Lambda^{(3N_c - N_f)/(N_c - N_f)} \tag{29.3.30}$$

and the m_i are the diagonal elements of the mass matrix that results when the original mass matrix is subjected to the $SU(N_f) \times \overline{SU(N_f)}$ transformation used to put the scalars in the form (29.3.26). The condition (27.4.10), that $f_{\text{total}}(q, \overline{q})$ should be stationary, has the solution

$$u_i^2 = \frac{1}{m_i} \left(\frac{\mathscr{K}}{N_c - N_f}\right)^{1 - N_f/N_c} \left(\prod_j m_j\right)^{-1/N_c} \tag{29.3.31}$$

Because we have put the scalar fields in a basis in which the u_i have a common phase, the m_i must also have a common phase in this basis. But the common phase of the u_i^2 is not unique — the $1/N_c$ powers in Eq. (29.3.31) tell us that the solution is undetermined by a factor $\exp(2i\pi n/N_c)$, with n an integer ranging from 0 to $N_c - 1$. (The two signs of u_i for a given u_i^2 are physically equivalent, because the whole theory is invariant under a non-anomalous symmetry with $q_{ai} \to e^{i\pi} q_{ai}$ and $\overline{q}_{ai} \to e^{-i\pi} \overline{q}_{ai}$.) The fact that there are N_c physically inequivalent solutions will turn up again in our discussion of the Witten index in the next section.

$C_1 = C_2$

This case is of some interest because, as we saw in Eq. (27.9.3), the simplest $N = 2$ supersymmetric Yang–Mills theory, when written in terms of $N = 1$ superfields, contains a single left-chiral superfield in the adjoint representation, for which** of course $C_2 = C_1$.

For $C_2 = C_1$ the function $\exp(2i\pi T)$ has $R = 0$, so that its appearance in $\mathscr{L}_\lambda^\sharp$ is not restricted by R invariance. The general form of the F-term

** Note that here C_2 refers to the representation of the gauge group furnished by the chiral superfields, so it is the same as the quantity C_2^b in Section 27.9, which refers to the representation furnished by complex scalars, but half of C_2^f, which refers to the representation furnished by all spinor fields, including the gauginos.